

Cosmological constraints using the density parameter of the Universe

Master's Thesis Defense

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1. Context & SNe Ia Datasets

- Λ CDM framework & H_0/ω_m tensions
- Pantheon+ and DES-SN5YR datasets
- Rigorous overlap removal (335 SNe)

2. Analysis Framework & Validation

- Method 1 : Hybrid Calibration
- Method 2 : Analytic H_0 Marginalisation
- Pipeline validation against baseline literature

3. Cosmological Results & DESI BAO

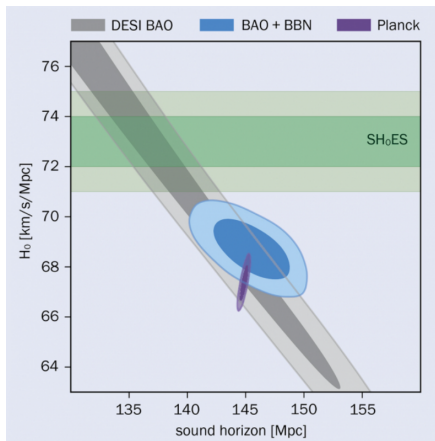
- SNe Ia constraints & physical ω_m tension
- SNe Ia + DESI DR2 BAO combination
- Restoration of consistency with Planck

4. Conclusions & Outlook

- Summary of baseline constraints
- Systematics vs. New Physics perspectives

1. Context & SNe Ia Datasets

Context of the study : The H_0 tension



Discrepancy between early-time (CMB) and late-time (SNe Ia) H_0 determinations. Source : CERN Courier (2021)

■ Late-time probes (Direct) :

- Distance ladder : SNe Ia + Cepheids
- $H_0 = 73.04 \pm 1.04 \text{ km.s}^{-1}.\text{Mpc}^{-1}$
(Riess et al. 2022, SH0ES)

■ Early-time probes (Model-dependent) :

- Cosmic Microwave Background (Λ CDM)
- $H_0 = 67.4 \pm 0.5 \text{ km.s}^{-1}.\text{Mpc}^{-1}$
(Planck Collab. 2018)

■ Status : Persistent 5σ tension on H_0 .

Cosmological Formalism : Λ CDM Model

Density Parameters

Critical density & fractions :

- $\rho_c(z) = \frac{3H^2(z)}{8\pi G}$
- $\Omega_i(z) = \frac{\rho_i(z)}{\rho_c(z)}$
- Physical density : $\omega_i = \Omega_{i,0} h^2$
($h = H_0/100 \text{ km s}^{-1}\text{Mpc}^{-1}$)

General closure relation :

$$\Omega_m + \Omega_r + \Omega_k + \Omega_\Lambda = 1$$

Flat Λ CDM Dynamics

Assumptions : $\Omega_k = 0$, $\Omega_r \approx 0$, $w = -1$

Expansion rate : $H(z) = H_0 E(z)$

With the dimensionless rate $E(z)$:

$$E(z) = \sqrt{\Omega_m(1+z)^3 + (1 - \Omega_m)}$$

→ Parameter constrained by SNe Ia luminosity distances $d_L(z)$.

Cosmological Formalism : Observables & Distances

Luminosity Distance $D_L(z)$

For a flat Λ CDM universe :

$$D_L(z) = \frac{c(1+z)}{H_0} \int_0^z \frac{dz'}{E(z')}$$

where $E(z) = \sqrt{\Omega_m(1+z)^3 + 1 - \Omega_m}$

Distance Modulus $\mu(z)$

Relates theoretical distance to observed magnitude :

$$\mu(z) = m_B - M$$

$$\mu(z) = 5 \log_{10} \left(\frac{D_L(z)}{1 \text{ Mpc}} \right) + 25$$

- m_B : Apparent magnitude in B -band
- M : Absolute magnitude of SNe Ia

Key Link for the study

The degeneracy between H_0 and M appears directly in $\mu(z)$ through $\frac{1}{H_0}$.

Extending the crisis : The Ω_m tension

- **In the Λ CDM framework** : Cosmological parameters are coupled. A discrepancy in H_0 propagates to the matter density parameter Ω_m .
- **Comparison of Ω_m constraints** :
 - **Type Ia Supernovae** : $\Omega_m = 0.334 \pm 0.018$ (Pantheon+, Brout et al. 2022)
 - **CMB (Planck)** : $\Omega_m = 0.315 \pm 0.007$ (Planck Collab. 2018)
- **Key Observation** :
 - *Planck* constrains the physical density $\omega_m = \Omega_m h^2$ with high precision.
 - Does the tension reside in H_0 , in Ω_m , or in the geometrical assumptions of Λ CDM ?

Motivation for this work

Re-evaluating the tension through physical density parameters (ω_m) using joint SNe Ia analyses (DES-SN5YR + Pantheon+).

Objectives of the Study

Main Goals

- **Re-constrain matter density parameters** : Derive updated constraints on Ω_m and the physical density $\omega_m = \Omega_m h^2$ using recent SNe Ia samples (*Pantheon+*, *DES-SN5YR*).
- **Quantify the Ω_m/ω_m tension** : Assess the statistical significance of the discrepancy in Ω_m/ω_m between SNe Ia datasets and *Planck* CMB measurements.
- **Joint cosmological analysis** : Combine SNe Ia constraints with early Baryon Acoustic Oscillations (BAO) data from the Dark Energy Spectroscopic Instrument (*DESI*).

Objects of Interest : Type Ia Supernovae (SNe Ia)

Physical Nature

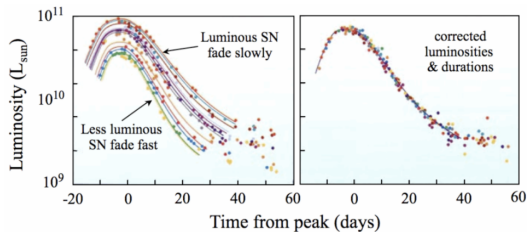
Thermonuclear explosion of a Carbon-Oxygen White Dwarf reaching the Chandrasekhar mass limit ($\sim 1.4M_{\odot}$) in a binary system.

Standardizable Candles

- **Peak luminosity** : Very bright ($M_B \approx -19.3$), detectable up to $z \sim 2.9$ (Pierel et al. 2024c).
- **Standardization (Tripp formula)** :

$$\mu_{\text{obs}} = m_B - M_0 + \alpha \cdot x_1 - \beta \cdot c + \Delta_M$$

- Low intrinsic scatter ($\sigma_{\mu} \approx 0.1$ mag) after Tripp calibration (x_1 : stretch, c : color).



Phillips relation : brighter SNe Ia decline more slowly (stretch correction x_1). Source : Roman et al.(2019)

Observational Datasets : DES-SN5YR vs Pantheon+

DES-SN5YR (Abbott et al. 2024)

- **1,829 SNe Ia** ($0.01 < z < 1.13$)
- Single, homogeneous survey
- High statistical power at mid/high- z

Pantheon+ (Brout et al. 2022)

- **1,701 SNe Ia** ($0.001 < z < 2.26$)
- Multi-survey compilation (SNLS, SDSS, HST, etc.)
- **42 Cepheid anchors** for H_0

Dataset Overlap & Error Propagation

- **Common objects** : 375 shared SNe Ia (consistency checks), including 40 duplicates in Pantheon+.
- **Covariance matrices** : Full $C = C_{\text{stat}} + C_{\text{syst}}$ (stat. + syst. errors).
- **Observable** : Fitted directly on distance moduli $\mu(z)$.

Complementary Probes : DESI BAO Data

DESI BAO Observables

Standard ruler from BAO sound horizon :

- **Transverse mode** : $D_M(z)/r_d$
(comoving angular distance)
- **Radial mode** : $D_H(z)/r_d = \frac{c}{H(z) r_d}$

Sound Horizon & Calibration

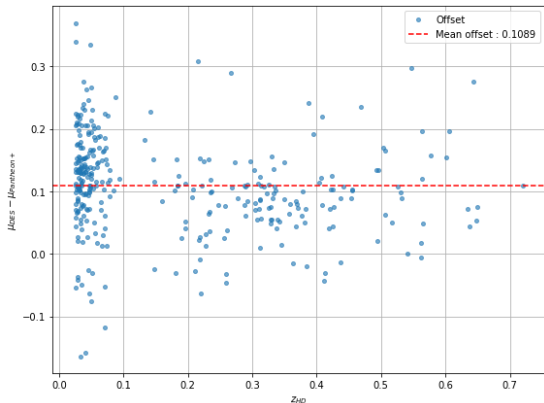
- **Sound horizon anchor** :
 $r_d = 147.05$ Mpc
(*Planck* 2018 Λ CDM baseline prior)
- **Tracers** : LRG, ELG, QSO, Ly- α
- **Covariance** : Diagonal errors across bins

Role in Joint Analysis

Combining SNe Ia (d_L) with DESI BAO ($D_M/r_d, D_H/r_d$) breaks parameter degeneracies between H_0 , Ω_m , and M .

2. Analysis Framework & Validation

Dataset Comparison : Common Objects Offset



Distance modulus residual $\Delta\mu(z)$ for shared SNe Ia.

Key Observations

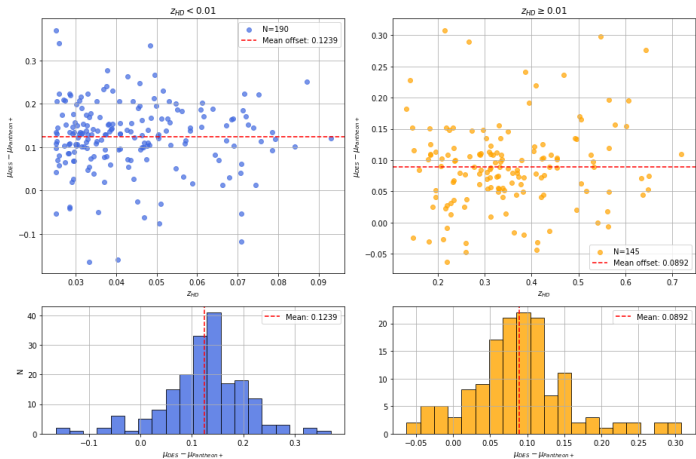
- **Sample overlap** : 335 shared SNe Ia at identical redshifts z .
- **Systematic shift** :

$$\langle \Delta\mu \rangle = 0.1089 \text{ mag}$$

Physical Origin

Discrepancy driven by differences in **photometric calibration**, **zero-points**, and **selection functions** between survey pipelines.

Low- z vs High- z Calibration Offset



Distance modulus residuals $\Delta\mu(z)$ split at $z = 0.1$.

Redshift Transition

Offset between local ($z < 0.1$) and Hubble flow ($z > 0.1$) SNe Ia :

$$\Delta\mu = 0.0347 \text{ mag}$$

Literature

- Consistent with **Efstathiou et al. (2024)** (≈ 0.04 mag).
- Points to low- z anchor calibration systematics.

Methods : χ^2 Minimisation & Likelihood

Joint χ^2 Formalism

$$\chi^2(\theta) = \Delta \mathbf{r}^T \mathbf{C}_{\text{stat}+\text{sys}}^{-1} \Delta \mathbf{r}$$

Residuals & Errors

- **Residual vector :**

$$\Delta r_i = \mu_{\text{obs}}(z_i) - \mu_{\text{th}}(z_i; \theta)$$

- **Covariance matrix $\mathbf{C}$:** Propagates both statistical and systematic uncertainties.

Inferred Parameters θ

- **Cosmology (Λ CDM) :** Ω_m, H_0 .
- **Nuisance parameter :** Absolute magnitude offset M_B .
- **Likelihood :**

$$\mathcal{L}(\theta) \propto \exp \left(-\frac{\chi^2(\theta)}{2} \right)$$

Methods : Marginalisation over H_0 / Zero-Point

Parameter Degeneracy in $\mu(z)$

The theoretical distance modulus separates cosmology (Ω_m) from normalization (H_0, M_B) :

$$\mu_{\text{th}}(z) = \underbrace{5 \log_{10} \left((1+z) \int_0^z \frac{dz'}{E(z'; \Omega_m)} \right)}_{\text{Cosmological shape } (\Omega_m)} + \underbrace{\mu_0}_{\text{Nuisance term } (H_0, M_B)}$$

where $\mu_0 = 25 + 5 \log_{10}(c/H_0)$ sets the absolute offset.

Marginalisation Integration

Integrate out the nuisance parameter μ_0 directly from the likelihood $\mathcal{L}$ (inspired by DES Collaboration (2024)) :

$$\mathcal{L}_{\text{marg}}(\Omega_m) = \int_{-\infty}^{+\infty} \mathcal{L}(\Omega_m, \mu_0) d\mu_0$$

(Executed analytically in the present case).

Key Advantages

- **Decouples Ω_m and H_0** : Measures matter density purely from light-curve shapes.
- **Systematics mitigation** : Unaffected by low- z zero-point or H_0 calibration offsets.

Methods : Joint Likelihood Analysis

Joint Likelihood Formalism

Assuming independent datasets, the combined likelihood multiplies (χ^2 sums) :

$$\chi_{\text{joint}}^2(\theta) = \chi_{\text{SNe}}^2(\theta) + \chi_{\text{DES}}^2(\theta)$$

SNe Ia Combination (DES / Pantheon+)

- **Overlap handling** : Duplicate removal (335 shared SNe) to prevent double-counting.
- **Zero-point calibration** : Accounted for via systematic covariance matrices.

SNe Ia + DESI BAO Synergy

- **Strict independence** : Zero cross-correlation between SNe Ia and DESI BAO.
- **Degeneracy breaking** : Drastically shrinks (Ω_m, H_0, M_B) confidence contours.

Methods : Sub-analyses Strategy

Method 1 : Hybrid Calibration

- **Pantheon+** : Fit both (Ω_m, H_0) anchored by Cepheids
($H_0 = 73.04 \pm 1.04$ km/s/Mpc, Riess+22).
- **DES-SN5YR** : Analytic marginalisation over $\kappa(H_0)$ (no local Cepheids).

Method 2 : Full Marginalisation

- **Both Datasets** : Analytic marginalisation over $\mu_0(H_0)$ for both samples.
- **Objective** : Direct comparison of Ω_m constraints independent of H_0 calibration.

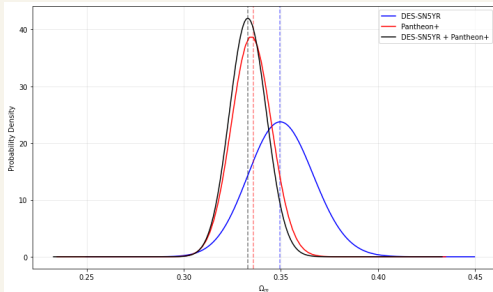
Common Objects Handling (Both Methods)

Shared SNe Ia are systematically removed from one dataset to avoid double-counting and preserve strict statistical independence.

3. Cosmological Results & DESI BAO

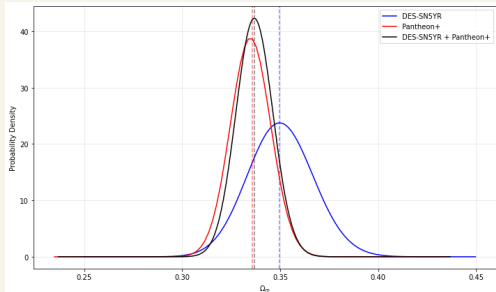
Results : Method 1 (Hybrid Calibration)

DES (- common) + Pantheon+



$$\Omega_{m,\text{joint}} = 0.333 \pm 0.009$$
$$\chi^2_{\text{red}} = 0.86$$

DES + Pantheon+ (- common)



$$\Omega_{m,\text{joint}} = 0.337 \pm 0.009$$
$$\chi^2_{\text{red}} = 0.89$$

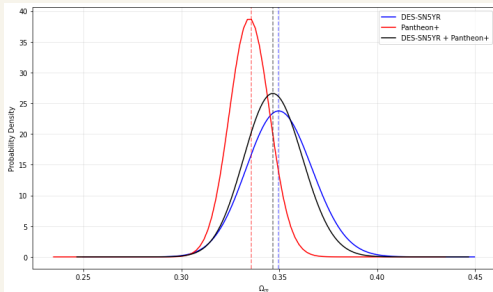
Impact of Overlap Removal

Both options yield fully consistent matter density values ($\Delta\Omega_m = 0.004 < 0.5\sigma$).

Removing overlap from either sample has negligible impact on constraints.

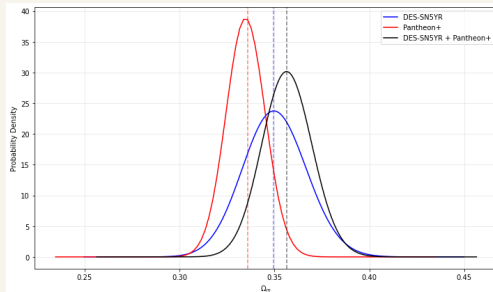
Results : Method 2 (Full Marginalisation)

DES (– common) + Pantheon+



$$\Omega_{m,\text{joint}} = 0.347 \pm 0.015$$

DES + Pantheon+ (– common)



$$\Omega_{m,\text{joint}} = 0.356 \pm 0.013$$

Robustness & Comparison with Method 1

Both options remain fully consistent ($\Delta\Omega_m = 0.009 < 0.5\sigma$).

Full marginalisation over H_0 slightly relaxes Ω_m constraints ($\approx \pm 0.014$ vs ± 0.009).

Final Results & Cosmological Tensions

Method 1 : Hybrid Calibration

$$\Omega_m = 0.335 \pm 0.009$$

(1.8 σ discrepancy with Planck)

$$\omega_m = \Omega_m h^2 = 0.179 \pm 0.007$$

(4.6 σ tension with Planck)

Method 2 : Full Marginalisation

$$\Omega_m = 0.352 \pm 0.014$$

(2.4 σ discrepancy with Planck)

$$\omega_m = \Omega_m h^2 = 0.188 \pm 0.009$$

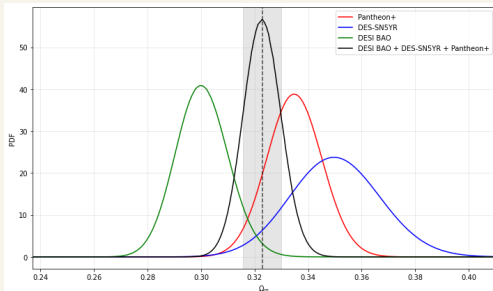
(4.7 σ tension with Planck)

Methodological Takeaway

- **Preferred pipeline :** Method 2 is preferred as it remains strictly H_0 -agnostic (no external Cepheid calibration required).
- **Physical tension :** Persistent $\geq 4.6\sigma$ tension on physical matter density ω_m against Planck CMB measurements regardless of method.

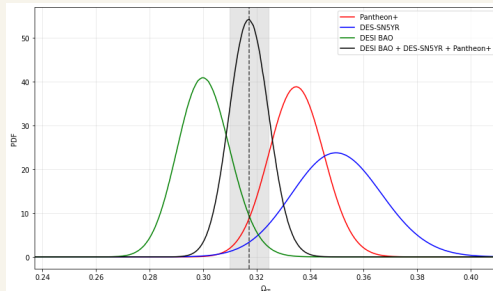
Results : SNe Ia + DESI BAO Joint Constraints

DES (- common) + Pan+ + DESI



$$\Omega_{m,\text{joint}} = 0.323 \pm 0.007$$

DES + Pan+ (- common) + DESI



$$\Omega_{m,\text{joint}} = 0.317 \pm 0.007$$

Impact of DESI BAO Combination

Adding BAO data breaks degeneracies and boosts precision ($\sigma_{\Omega_m} \rightarrow \pm 0.007$).

Constraints remain fully robust against overlap removal choice ($\Delta\Omega_m = 0.006 < 1\sigma$).

Final Synthesis : SNe Ia + DESI BAO Constraints

Combined SNe Ia + BAO Baseline (Method 2)

$$\Omega_m = 0.320 \pm 0.007$$

(Averaged over overlap removal options : 0.323 vs 0.317)

Impact of BAO Combination

- **High Precision** : Error bars reduced to ± 0.007 ($\approx 2\times$ tighter than SNe alone).
- **Degeneracy Breaking** : Effectively breaks the $\Omega_m - H_0$ parameter correlation.

Planck CMB Compatibility

- **Full Agreement** : Perfectly consistent with Planck Λ CDM ($\Omega_m = 0.315 \pm 0.007$).
- **Calibration Independent** : Holds strictly without assuming any local H_0 anchor.

4. Conclusions & Outlooks

Conclusions & Perspectives

Main Cosmological Findings

- **Robust Ω_m constraints** : Perfectly consistent across overlap removal methods ($\Omega_m = 0.320 \pm 0.007$ with DESI).
- **Persistent ω_m tension** : Significant tension ($\geq 4.6\sigma$) on physical matter density using SNe Ia alone vs Planck.

Physical Interpretation

- **SNe + BAO synergy** : Combining SNe Ia with BAO fully restores agreement with Planck within flat Λ CDM.
- **Calibration impact** : Tension is primarily driven by H_0/M_B absolute zero-point calibration, not light-curve shapes.

Perspectives : Systematics vs. New Physics

- **Systematics** : Need for deeper investigation into low- z zero-point calibration and local flow corrections.
- **Beyond Λ CDM** : Residual tensions may signal dynamical dark energy ($w(z)$ evolution) or modified gravity, to be tested with upcoming DESI DRs and Euclid.

Acknowledgements

Scientific Supervision & Team

- **Pr. Alain BLANCHARD** : Sincere gratitude for continuous guidance, scientific insight, and support throughout this project.
- **Cosmology Group** : Thanks to the entire team for fruitful discussions and valuable expertise.

Host Institution & Colleagues

- **Host Laboratory** : Institut de Recherche en Astrophysique et Planétologie (**IRAP** / **OMP**).
- **Fellow Interns** : Warm thanks to the intern group for the great working atmosphere and mutual support.

Thank You !

Thank you for your attention.
I am open to any questions from the jury.

5. Appendices

Appendix : Redshift Definitions & Corrections

Redshift Definitions in Supernova Datasets

Modern SNe Ia pipelines (Pantheon+, DES) distinguish three redshift measurements :

- z_{HEL} (**Heliocentric**) : Directly observed spectroscopic redshift in the Solar System frame.
- z_{CMB} (**CMB frame**) : Corrected for the motion of the Solar System relative to the CMB dipole.
- z_{HD} (**Hubble Diagram**) : Corrected for local coherent peculiar velocity fields (v_{pec}).

Physical Application

- **Distance modulus** $\mu(z_{\text{HD}})$: z_{HD} is used for cosmological expansion to minimize peculiar flow biases.
- **Flux dilation** $(1 + z_{\text{HEL}})$: z_{HEL} accounts for time dilation in light curves.

Impact on Systematics

- **Low- z sensitivity** : v_{pec} corrections ($\approx 150\text{--}250$ km/s) dominate the uncertainty budget for $z < 0.05$.
- **Direct link to H_0** : Mismodeling local flows introduces a direct shift on H_0 zero-point calibration.

Appendix : Uncertainty Budget & Covariance Matrix

Total Covariance Formalism

In supernova cosmology, the likelihood $\chi^2 = \Delta\mu^T \mathbf{C}^{-1} \Delta\mu$ incorporates the full error budget :

$$\mathbf{C} = \mathbf{C}_{\text{stat}} + \mathbf{C}_{\text{sys}}$$

where $\mathbf{C}_{\text{stat}}$ is strictly diagonal and $\mathbf{C}_{\text{sys}}$ captures off-diagonal correlations.

Statistical Uncertainties ($\mathbf{C}_{\text{stat}}$)

- **Sources** : Flux measurement noise, Poisson photon noise, intrinsic dispersion ($\sigma_{\text{int}} \approx 0.10\text{--}0.14$ mag).
- **Property** : Uncorrelated ; scales down with sample size ($\propto 1/\sqrt{N}$).

Systematic Uncertainties ($\mathbf{C}_{\text{sys}}$)

- **Sources** : Photometric zero-point calibrations, SALT3 light-curve model, host-galaxy mass step, dust extinction (R_V).
- **Property** : Introduces non-zero off-diagonal terms ; irreducible by sample size alone.

Appendix : Analytic Marginalisation (1/2) — Setup

Physical Meaning of the Offset κ

The nuisance parameter κ absorbs both H_0 and the absolute magnitude M_B :

$$\mu_0 = 5 \log_{10} \left(\frac{c/H_0}{1 \text{ Mpc}} \right) + M_B + 25$$

Matrix Formulation

Defining Hubble residuals $\Delta \mathbf{r}(\Omega_m) \equiv \mu_{\text{obs}} - \mu_{\text{model}}(\Omega_m)$, the total χ^2 is :

$$\chi^2(\Omega_m, \mu_0) = (\Delta \mathbf{r} - \mu_0 \mathbf{1})^T \mathbf{C}^{-1} (\Delta \mathbf{r} - \mu_0 \mathbf{1})$$

where $\mathbf{C}$ is the full statistical + systematic covariance matrix.

Appendix : Analytic Marginalisation (2/2) — Derivation

Analytical Minimisation over κ

Solving $\frac{\partial \chi^2}{\partial \mu_0} = 0$ gives the optimal zero-point shift :

$$\mu_{0,\text{opt}} = \frac{\mathbf{1}^T \mathbf{C}^{-1} \Delta \mathbf{r}}{\mathbf{1}^T \mathbf{C}^{-1} \mathbf{1}}$$

Marginalised χ^2 Statistic

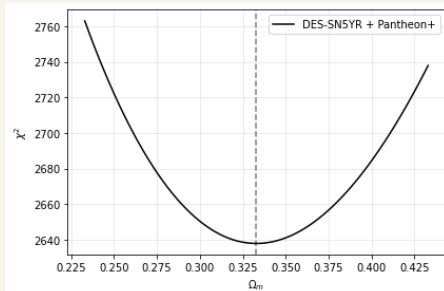
Substituting $\mu_{0,\text{opt}}$ back into the χ^2 profile yields :

$$\chi_{\text{marg}}^2(\Omega_m) = \Delta \mathbf{r}^T \mathbf{C}^{-1} \Delta \mathbf{r} - \frac{(\mathbf{1}^T \mathbf{C}^{-1} \Delta \mathbf{r})^2}{\mathbf{1}^T \mathbf{C}^{-1} \mathbf{1}}$$

Constrains Ω_m strictly from light-curve relative shapes, independent of H_0/M_B .

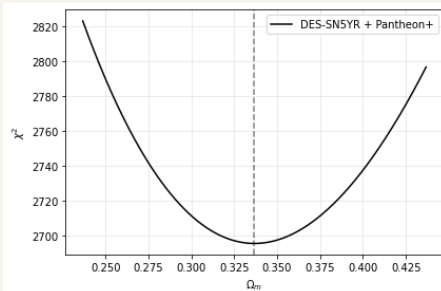
Appendix : $\chi^2(\Omega_m)$ Profile Likelihoods

DES-SN5YR (- common) + Pantheon+



$$\chi^2_{\min} = 2638$$

DES-SN5YR + Pantheon+ (- common)



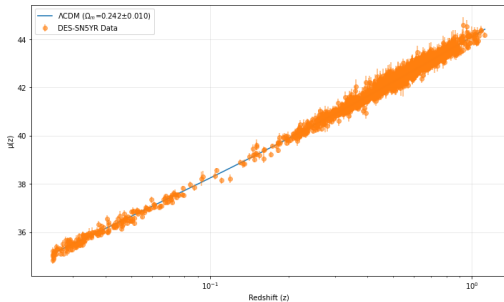
$$\chi^2_{\min} = 2695$$

Profile Shape & Curvature Analysis

Both profiles display identical parabolic shapes around the minimum ($\Delta\chi^2$ curvature).
The vertical offset ($\Delta\chi^2_{\min} = 57$) is purely driven by different total sample sizes (N_{dof}).

Appendix : Pipeline Validation — Pantheon+ $\mu(z)$ in DES-SN5YR

DES-SN5YR Dataset Sensitivity Test



Impact of Replacing Shared $\mu(z)$

Cross-calibration test on common SNe Ia :

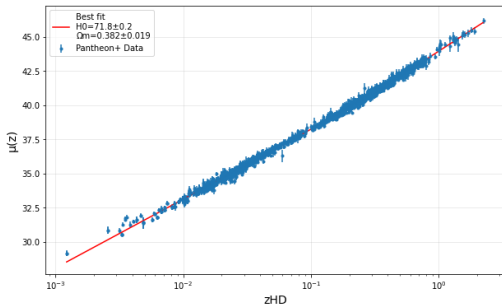
- **Original DES-SN5YR (2024) :**
 $\Omega_m = 0.352 \pm 0.017$ (marginalised over H_0)
- **Substituted ($\mu_{\text{DES}} \rightarrow \mu_{\text{Pan+}}$) :**
 $\Omega_m = 0.242 \pm 0.010$

Physical Interpretation

Injecting Pantheon+ distance moduli for common SNe Ia induces a massive shift ($\Delta\Omega_m \approx -0.11$), proving strong cross-calibration / zero-point differences between the two pipelines.

Appendix : Pipeline Validation — DES-SN5YR $\mu(z)$ in Pantheon+

Pantheon+ Dataset Sensitivity Test



Impact of Replacing Shared $\mu(z)$

Reciprocal test on Pantheon+ baseline :

- **Original Pantheon+ (2022) :**

$$\Omega_m = 0.334 \pm 0.018$$

$$H_0 = 73.6 \pm 1.1 \text{ km/s/Mpc}$$

- **Substituted ($\mu_{\text{Pan}+} \rightarrow \mu_{\text{DES}}$) :**

$$\Omega_m = 0.382 \pm 0.019$$

$$H_0 = 71.8 \pm 0.2 \text{ km/s/Mpc}$$

Physical Interpretation

Symmetrically drives Ω_m higher ($\Delta\Omega_m \approx +0.05$) and reduces H_0 , confirming that shared object zero-points must be strictly unified or isolated before joint fitting.